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RESEARCH ARTICLE

A Novel Integrated Approach for Stock Prediction Based on Modal Decomposition Technology and Machine Learning

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ABSTRACT After the COVID-19 ended, the global economy gradually recovered. Due to the nonlinearity, complexity, and high noise of financial time series, stock price prediction has become one of the most challenging tasks in the stock market. To tackle this challenge and enhance the prediction performance in the complicated stock markets, we propose a novel integrated approach based on Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), Long Short-Term Memory (LSTM), and ensemble learning algorithm LightGBM to simultaneously improve the fitting and accuracy of stock price prediction. In addition, to prevent overfitting and improve predictive performance, this study adopted the Simulated Annealing (SA) algorithm for optimization. The predictive performance of the proposed hybrid model is comprehensively evaluated by comparing it with single LSTM, RNN, and other popular hybrid models. Three evaluation metrics, namely Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and accuracy, are used to compare the aforementioned models. The experimental results indicate that the proposed hybrid CEEMDAN-LSTM-SA-LightGBM model outperforms all other comparative models in this study with better fitting and accuracy.

INDEX TERMS LSTM, CEEMDAN, simulated annealing algorithm, stock price forecasting, LightGBM.

I. INTRODUCTION

The stock market plays a great role in the capital market, which can promote capital flow, optimize asset allocation, and stimulate better and faster economic development. At the same time, the stock market is a remarkable place for investors to invest and a focal point for the state to regulate economic trends. Investors are more concerned about how to maximize profits while minimizing risks, and the state is always alert to the occurrence of economic crises. All these urgently need to make accurate judgments on the stock price trend. With the continuous development of the Internet in the financial industry, the stock market has generated a large amount of network data. Due to the sensitivity of the stock market to various factors, it is difficult to draw accurate and valid conclusions from these massive stored data. In addition, financial time series are characterized by instability, nonlinearity, and noise [1]. Therefore, in the face of the stock market with complicated information and ever-changing status, how to see the essence through the complex phenomenon, grasp the market dynamics and asset price trends, and avoid risks are the core issues that stock investors and the government

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attach great importance to and study in depth. Besides, the stock market is also an exceptionally complex nonlinear dynamic system. Many internal and external factors affect its volatility, most of which are difficult to quantify [2]. Meanwhile, the amount of information and data related to stocks is very large, making traditional non-artificial intelligence methods often unsatisfactory in predicting stock prices [3].

To address these challenges, advanced computational techniques, including Machine Learning (ML) and Deep Learning (DL), have been increasingly adopted in recent years [4]. LSTM networks are particularly effective in modeling the temporal dynamics of stock prices, making them a preferred choice for financial time series prediction [5]. By utilizing CEEMDAN to preprocess and decompose original stock price data, the LSTM network can focus on predicting each IMF individually, thus enhancing prediction accuracy and robustness. This hybrid method capitalizes on the strengths of both techniques, providing a comprehensive framework for analyzing and forecasting stock market trends [6]. Moreover, ensemble learning algorithms, such as Random Forests and Gradient Boosting Machines, have been incorporated to further refine prediction models. These techniques combine the predictions of multiple base models to improve generalizability and reduce the risk of overfitting [7]. The integration of ensemble methods with the CEEMDAN-LSTM framework promises to yield superior predictive performance, as evidenced by recent studies [8], [9].

This research aims to propose a high-precision and high-fitting stock prediction modeling method, which provides practitioners with more effective tools to grasp the dynamic trends of the stock market, avoid market risks, and enhance excess return ability. This approach is based on modal decomposition technology and machine learning. The CEEMDAN is adopted for suppressing noise and reducing errors, and the LSTM neural network for long dependencies, as well as optimized LightGBM which is a highly efficient ensemble learning algorithm for improving the stability of the prediction model. The main innovation in this research lies in:

(1) Applying the CEEMDAN method with adaptive decomposition capability to the modeling process of stock price prediction, to obtain relatively simple and independent components and trend terms of volatility characteristics, avoids the technical challenge of directly extracting volatility patterns from time series data utilizing existing modeling methods and greatly reduces the difficulty of time series prediction modeling.

(2) Utilizing LSTM to construct a prediction model for various feature sequences overcomes the limitations of traditional statistical modeling methods, which are too strict in assuming the classification of data distribution. It can also mine long-term dynamic dependencies contained in complex financial time series more efficiently.

(3) An integrated algorithm prediction method is proposed, which combines the strengths of four outstanding fundamental algorithms and reduces computation through feature extraction. By analyzing six stocks across 10 comparative models, the prediction model proposed in this paper exhibits both fitting and accuracy.

The main arrangements of this paper are as follows:

Literature review in section II. This section reviews existing studies on hybrid stock prediction methods based on modal decomposition technologies and machine learning, highlighting the strengths and challenges of previous approaches.

Section III introduces each algorithm adopted in this study, including CEEMDAN, LSTM, LightGBM, and simulated annealing algorithm separately.

Section IV describes the modeling process. The proposed CEEMDAN-LSTM-SA-LightGBM model is described in detail. This section outlines the experimental design, including the dataset description, data preprocessing steps, and the implementation details of model building.

Section V explains the analysis of experimental results and compares the performance of the proposed model with the comparative models by utilizing evaluation indicators.

Section VI is the conclusion of the paper. This section summarizes the main findings of this study, and proposes the challenges and limitations encountered in this study.

II. PREVIOUS LITERATURE

There are mainly two categories of forecasting stock: linear prediction models and nonlinear prediction models [10]. Traditional statistical methods belong to linear prediction models, for example, Autoregressive Integrated Moving Average Model (ARIMA) and moving average model. The application of these time series models has made certain progress in financial market forecasting research [11], [12], [13]. In recent decades, artificial intelligence and big data technology have continuously developed, more and more nonlinear prediction models have emerged, such as decision trees [14], support vector machines [15], logistic regression [16], Convolutional Neural Network (CNN) [17], Recurrent Neural Network (RNN) [18], and Long Short-Term Memory (LSTM) [19] for stock prediction. Among them, LSTM possesses the capability to tackle crucial issues with long-term dependencies in time series, addressing the drawback of RNNs that tend to forget prior state information over time [20]. In recent years, LSTM models have shown significant advantages in time series prediction [21], [22], [23].

On the basis of existing research, stock price prediction has been an undeniable challenge that needs to be further explored. With the deepening research on the microstructure of financial markets and the psychology of trading behavior, scholars have gradually realized that there is still room for improvement in the prediction performance of intelligent prediction models, despite the improvement in stock prediction performance compared with single algorithmic models [24]. As pointed out by Peters [25], the financial market is composed of traders with different investment time levels, such as short-term, medium-term, and long-term. Accordingly,

different types of traders have different investment philosophies that are dispersed and reflected on different time scales. Different traders also view and influence the market from different time scales. Therefore, studying the effective forecasting of the stock market at different time scales from a multiscale perspective may lead to new findings. In addition, the difficulty of forecasting financial time series increases due to the presence of noise in the financial time series [26]. Therefore, scholars have begun to explore a better performance stock prediction method, which is a synthesis prediction model based on mode decomposition algorithms, the original signals at different scales are decomposed to reduce noise. TABLE 1 presents a comparative analysis of representative financial time series prediction methods that utilize modal decomposition techniques and machine learning algorithms over the past three years.

As evident from the literature mentioned, scholars are dedicated to exploring the application of modal decomposition technology in predicting stock prices. The literature confirms the feasibility of combining modal decomposition technology with machine learning in financial time series and produces satisfactory prediction results. However, the literature primarily focuses on predicting stock index, without a thorough investigation into predicting stock volatility. Furthermore, the evaluation criteria for predictions primarily focus on the predictive fit, overlooking the precision of predicting stock volatility, and due to the modal decomposition technique decomposing the original signal into multiple IMFs, it generally increases computational overhead.

Additionally, ensemble learning is a very important model construction concept in the field of machine learning. It can improve the generalization ability of predictions effectively. Nowadays, ensemble learning is known as the primary research direction of current machine learning gradually and becoming a research hotspot in the field of machine learning [34], and the algorithms are extensively applied in predictive applications [35], [36], but few scholars have further explored stock prediction models based on modal decomposition techniques and ensemble learning algorithms in the current research literature.

Therefore, building on previous research, this study combines the strengths of CEEMDAN, LSTM, the ensemble learning algorithm LightGBM, and the optimization technique to form a novel comprehensive stock prediction approach for future exploration and validation.

III. METHODS

Section III introduces each algorithm adopted in this study, including CEEMDAN for decomposing original signals and removing noise, the LSTM model for building test sets, the LightGBM model for stable prediction, and Simulated Annealing (SA) optimization algorithm.

A. CEEMDAN

Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) is an advanced version of modal decomposition technology. Empirical Modal Decomposition (EMD) is a classical frequency-domain signal processing algorithm with significant advantages in analyzing nonlinear and nonstationary signal sequences [37]. EMD has achieved significant results in signal decomposition, but modal aliasing often occurs during its operation. Then, Wu and Huang [38] proposed EEMD by adding white noise with a normal distribution to the original sequence as an improved method, this method enables the sequence to have continuity at different scales, avoiding the modal aliasing defect caused by the discontinuity of IMF in EMD, which affects the subsequent processing and analysis of signals. However, EEMD can easily lead to a decrease in decomposition efficiency and significant reconstruction errors. Yeh et al. [39] proposed a new complementary EEMD (CEEMD) method that effectively eliminates the added white noise and reduces errors caused by reconstructed data during the ensemble averaging process through the method of adding a pair of standard white noise (positive and negative) of the same amplitude. In the second year, Torres et al. [40] added multiple sets of independent and identically distributed Gaussian white noise to the original sequence data, then performed EMD decomposition to obtain IMF components and residuals. This avoids adding white noise every time EEMD decomposes, achieves an adaptive noise addition process, improves resolution efficiency, reduces residual noise, and reduces the reconstruction error of the original sequence data. The specific algorithm process of CEEMDAN will follow the following steps:

Step 1: Generate noisy data sets:

$$x_i(t) = x(t) + \varepsilon_0 w_i(t) \tag{1}$$

In the above equation, $x(t)$ is the original time series data, $w_i(t)$ (i = 1, 2…T) is Gaussian white noise that follows a normal distribution, and $\boldsymbol{\varepsilon}$ is the standard deviation of the noise;

Step 2: Operate the EMD method to decompose the sequence data, gain the first-order modal component IMF of each sample signal, and calculate the mean of each IMF mentioned above, use this mean as the first IMF of $x(t)$, the next work is to calculate the residual $\boldsymbol{r_1}(\boldsymbol{t})$ for the first stage, where $EMD_1(\cdot)$ represents the first IMF obtained by EMD:

$$IMF_1(t) = \frac{1}{m}\sum\nolimits_1^m IMF_1^i(t) = \frac{1}{m}EMD_1(x_i(t)) \tag{2}$$

$$r_1(t) = x(t) - IMF_1(t) \tag{3}$$

Step 3: Continue to perform EMD to obtain the components and obtain the mean, obtaining the j-th IMF and the residual rj(t) after the decomposition. The equations are as follows:

$$IMF_j(t) = \frac{1}{m}\sum\nolimits_1^m EMD_1(r_{j-1}(t) + \varepsilon_{j-1}EMD_{j-1}(w_i(t))) \tag{4}$$

$$r_j(t) = r_{j-1}(t) - IMF_j(t) \tag{5}$$

TABLE 1. Financial time series prediction methods based on modal decomposition techniques and machine learning algorithms over the past three years.

Reference	Year	Method	Evaluation Metrics	Value Predicted	Advantage	Challenges
[27]	2021	CEEMDAN-LSTM	RMSE, MAE,MAPE,MSE	Stock Index	Improves the fitting and robustness of prediction model	High computational cost
[28]	2022	SVR with MEMD	MAPE, RMAE, DS	Stock Index	Effective in handling non-stationary data and capturing nonlinear relationships	Computationally intensive and requires careful parameter tuning
[29]	2023	EMD-LSTM	RMSE, MAE, MAPE	Stock Index	Combines the strengths of EMD and LSTM, providing improved prediction fit	Complexity in model implementation and high computational cost
[30]	2023	AdaBoost-AAFSA-Elman with CEEMDAN	RMSE, MAE, MAPE	stock closing price	Incorporates multiple models and algorithms for enhanced prediction performance	Requires extensive data preprocessing
[31]	2023	EEMD-Hurst-LSTM	RMSE, MSE, R2	Commodity Futures	Improves forecasting performance by integrating EEMD and LSTM with Hurst exponent analysis	Involves complex calculations
[32]	2024	CEEMDAN-ARIMAX	RMSE, MAE	Stock Index	Provides better decomposition and noise reduction for financial market forecasting	High computational complexity and computational resources
[33]	2024	2LE-CEEMDAN	MAE, RMSE, MSE, R2	Stock Index	Offers enhanced Denoising function for more accurate time series analysis	High computational complexity and requires extensive parameter tuning

Step 4: Finally, when the residual does not exceed two extreme points and cannot continue the decomposition work, the CEEMDAN algorithm will be stop. The residual trend is obvious and direct, and decomposes the original signal sequence into m modal components and residual term $R(t)$, the original signal $f(t)$ has the relationship described by the equation provided below:

$$f(t) = \sum_{j=1}^{m} IMF_j(t) + R(t) \tag{6}$$

B. LSTM

Long Short-Term Memory (LSTM) neural network [41] has been favored by scholars since its proposal due to its excellent performance in both long and short-term memory. The innovation of the LSTM model lies in its introduction of gate mechanism. It contains input gate, forget gate, and output gate. These gates are used for controlling the flow of information. Among them, the input gate is mainly used to control the inflow of new input information. The forget gate is mainly used to control the outflow of previous state information. The output gate is mainly used to control the output of current state information. Due to the ordered operation of these three gates, the LSTM network has outstanding predictive performance in various fields. Especially, there are certain advantages of LSTM in dealing with long-term dependency problems [42]. The schematic diagram of the LSTM network is shown in Fig. 1.

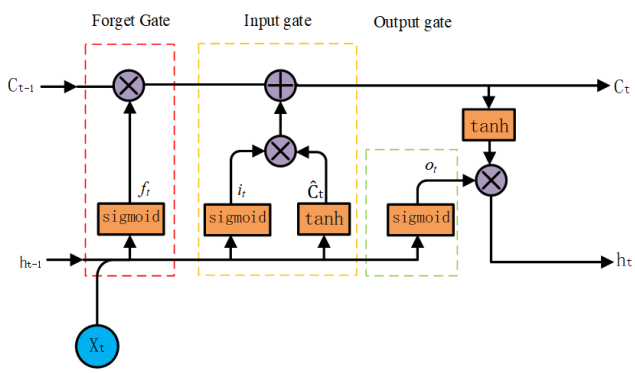

FIGURE 1. Structure diagram of LSTM model.

The following equation describes the workflow of the LSTM network.

$$i_t = \sigma\left(W_i * [h_{t-1}, X_t] + b_i\right) \tag{7}$$

$$f_t = \sigma\left(W_f * [h_{t-1}, X_t] + b_f\right) \tag{8}$$

$$o_t = \sigma\left(W_o * [h_{t-1}, X_t] + b_o\right) \tag{9}$$

$$\hat{C}_t = tanh\left(W_c * [h_{t-1}, X_t] + b_c\right) \tag{10}$$

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t \tag{11}$$

$$h_t = o_t * \tanh\left(C_t\right) \tag{12}$$

Among them, X_t, h_t, and C_t stand for the input, output, and state vectors of the cell, respectively; f_t, i_t and o_t mean forget gate output, input gate output, and output gate output vectors, respectively; W and b represent weight vectors and bias terms.

Due to the efficient coordination between the three gates of the LSTM network, it is particularly well suited for forecasting complex financial time series. During model training, LSTM updates its parameters adopting the backpropagation algorithm, enabling the model to adapt to various input sequences.

C. LIGHTGBM

As a highly efficient gradient boosting tree framework, Light Gradient Boosting Machine (LightGBM) is developed by the Microsoft team in 2017 [43]. The innovation of this algorithm lies in the introduction of two new technologies based on traditional GBDT. They are Gradient-based One Side Sampling(GOSS) and Exclusive Feature Bundling (EFB).

The function of GOSS is to filter the data and only use a small amount of data. The improvement of the GOSS algorithm lies in sorting gradients, selecting samples with larger gradients by percentage, and randomly selecting some samples with smaller gradients, compensating for a constant, as input data for child nodes, it greatly reduces the amount of training data. EFB technology achieves bundling of exclusive features to reduce the number of features. EFB algorithm converts the filtering problem of mutually exclusive features into a coloring problem of the graph [44], and the mutually exclusive features are converted into points of the same color, sorted according to the degree of the mutually exclusive bundle, and then merged with the mutually exclusive features within the tolerance range.

Furthermore, LightGBM also adopts a decision tree algorithm based on histogram as shown in Fig. 2, which saves half of the execution time of the GBDT algorithm, and its leaf growth strategy with maximum depth constraint makes the overall execution of the algorithm more efficient. Compared with the traditional Gradient Boosting Decision Tree (GBDT) algorithm, the LightGBM algorithm offers significant advantages in terms of training speed and accuracy [45]. It supports parallel training, suitable for training tasks on large-scale datasets. Besides, it has the advantages of flexible hyper-parameter adjustment and interpretability [46], [47].

D. SIMULATED ANNEALING

As a compact and robust simulation algorithm, Simulated Annealing (SA) can provide outstanding settlement for single-objective and multi-objective optimization, it can also greatly reduce computational time [48]. SA algorithm has a certain probability of accepting a worse solution as a new solution after each jump, so it has strong climbing ability and is not easily trapped in local optima [49]. Simulated annealing not only finds the global optimum, but because it is a strongly robust algorithm, it is unlikely to invalidate the conditioning function [50].

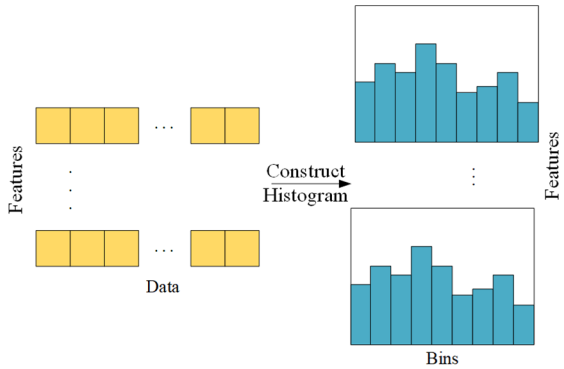

FIGURE 2. Schematic diagram of the histogram algorithm.

SA algorithm is based on statistical physics and incorporates the natural mechanisms of the crystal annealing process in physical systems. The algorithm adopts the Metropolis criterion to accept the generated optimal problem solution. The key to the SA algorithm is to reject the local minimum problem solution with a certain probability, as the number of iterations increases, the temperature T gradually decreases, and then jumps out of the local extreme point to continue mining other state solutions in the state space, thereby obtaining the global optimal problem solution. Equation (13)-(14) are described as the process of simulated annealing, where P represents the probability of rejection.

$$\mathrm{P} = \begin{cases} 1 & \text{if } f\left(x'\right) < f(x) \\ \exp\left(-\dfrac{f\left(x'\right) - f(x)}{T}\right) & \text{if } f\left(x'\right) > f(x) \end{cases} \tag{13}$$

$$T_{k+1} = \alpha T_K \quad (0 < \alpha < 1) \tag{14}$$

The simulated annealing algorithm excels in optimizing the hyper-parameters of machine learning models and enhances model accuracy effectively. In this study, The SA optimization algorithm is used for hyper-parameter selection in the LightGBM model. Cross validation is used for evaluating the quality of hyper-parameters, with CV=5 and scoring selecting Normalized Mean Square Error (NMSE) is used as evaluation indicator.

IV. MODEL BUILDING

This section will first analyze the historical data of stocks, then introduce the process of decomposing original signals and extracting effective features to reconstruct the dataset, as well as the training and prediction process. The process of model building is shown in Fig. 3.

A. DATA

The simulation platform of this research is conducted by Pycharm and programming is implemented by Python language. The experimental data is downloaded from the Kaggle

competition platform (https://www.kaggle.com/datasets/kane6543/ most-watched-stocks-of-past-decade20132023). To test and verify the universality of the stock forecasting method, historical stock data from different sectors such as technology, services, and entertainment are selected for analysis and validation. Six stock time series historical datasets including "Salesforce", "Ali baba", "Walt Disney", "Electronic Arts", "Netflix", and "AMD" are utilized for this study. The time range for these selected stocks is from January 2013 to December 2022. Historical trends of six stocks are shown in Fig. 4, and the descriptive statistical results of each stock are shown in TABLE 2.

B. CEEMDAN DECOMPOSITION

Based on descriptive statistics of stocks above, all six stock trends exhibit typical nonlinear and unstable financial time series data characteristics with some noise. To accurately predict these complex financial time series, the CEEMDAN algorithm will be used to decompose the original signals, to achieve the goal of removing noise in the next step of research work.

According to the mathematical theories mentioned in Section III, the CEEMDAN algorithm is adopted to decompose the historical feature data of six stocks. The principle of decomposition is nearly the same, so this paper only shows the decomposition results of "Salesforce" stock, as shown in Fig. 5.

As can be seen from the decomposition result, the original features including "Close", "High", "Low", "Open", "Adj Close", "Volume" and "Close-Open" (today's closing price minus today's opening price) are decomposed by the CEEMDAN algorithm, resulting in a total of 67 IMFs and residuals. The decomposed signal exhibits increased smoothness and regularity. However, the sheer number of decomposed signals poses a significant computational cost when used directly in calculations. Additionally, including an excessive number of feature values can adversely affect the prediction outcomes. Consequently, the subsequent step involves extracting feature values and reconstructing the dataset.

C. BUILDING DATASET

For comprehensively and effectively extracting feature values, the CEEMDAN algorithm is adopted for signal decomposition of the original feature values. As different data features have different dimensions, normalization processing is required, equation (15) describes the calculation process, Where x_{max} and x_{min} represent the maximum and minimum values in the sequence, respectively.

$$\mathrm{x} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{15}$$

After standardizing the features, the next step is to extract features by calculating the Pearson correlation coefficient between each decomposed dimension and the stock's "rise and fall" attributes. The method of utilizing 50% as the correlation threshold to screen features can retain the most representative and predictive features while avoiding overfitting and high model complexity issues [51], [52]. Therefore, in this study, 50% of the correlation coefficient will be selected as the criterion for feature selection. The process of extracting features is shown in Fig. 6. Taking "Salesforce" stock as an example, after decomposition, 67 components are generated. However, after 50% correlation screening, only these 3 feature values including "IMF-Close1", "IMF-Adj Close1", and "IMF_close-open1" are retained as the final feature variables. This method can significantly reduce input variables, thereby reducing runtime and computational costs.

When feature selection is completed, subsequent steps involve creating a new dataset arranged in chronological order. The training and test sets have a ratio of 80% and 20%, respectively. To establish a reliable stock prediction approach, this study added a working process of adjusting the dataset, which involves utilizing a double-layer LSTM network to reconstruct a new test set. Each attribute in the new dataset is trained and predicted utilizing the double-layer LSTM. This research adopts the rolling prediction modeling method and sets the sliding window size of each attribute to be 5, sets epochs to be 10, and set batch_size to be 25. The structure of proposed LSTM is shown in Fig. 7. To ensure the global minimum loss function when the model converges quickly, the optimizer is selected as Adagrad. The component prediction result after decomposing the "Close" attribute adopting double-layer LSTM network in the new dataset of "Salesforce" stock is shown in Fig. 8.

From the predictive performance of the double-layer LSTM, it exhibits excellent predictive fitting capabilities, indicating the feasibility of utilizing the double-layer LSTM model for prediction. Subsequently, the feature-extracted attribute values are individually trained and predicted through the partitioned dataset by the LSTM model, and the prediction results of each attribute are saved as a new test set.

D. MODELING

Based on the excellent prediction performance of the LightGBM model introduced in Section III, LightGBM will be used to predict stock prices in this study. Meanwhile, there are several hyperparameters in the LightGBM model, and the selection of hyperparameters will have an important impact on the prediction performance. Therefore, as recommended in Section III, this research will use simulated annealing algorithm to optimize the hyper-parameters in the LightGBM model to further improve the predictive performance of LightGBM.

Essentially, in this method, during the training process of the LightGBM model, due to the influence of the core parameters in the LightGBM model on the quality of model training, the simulated annealing algorithm is employed to optimize the "n_estimators", "max_depth", and "learning_rate" parameters in the LightGBM model. The specific optimization process is shown in Fig. 9.

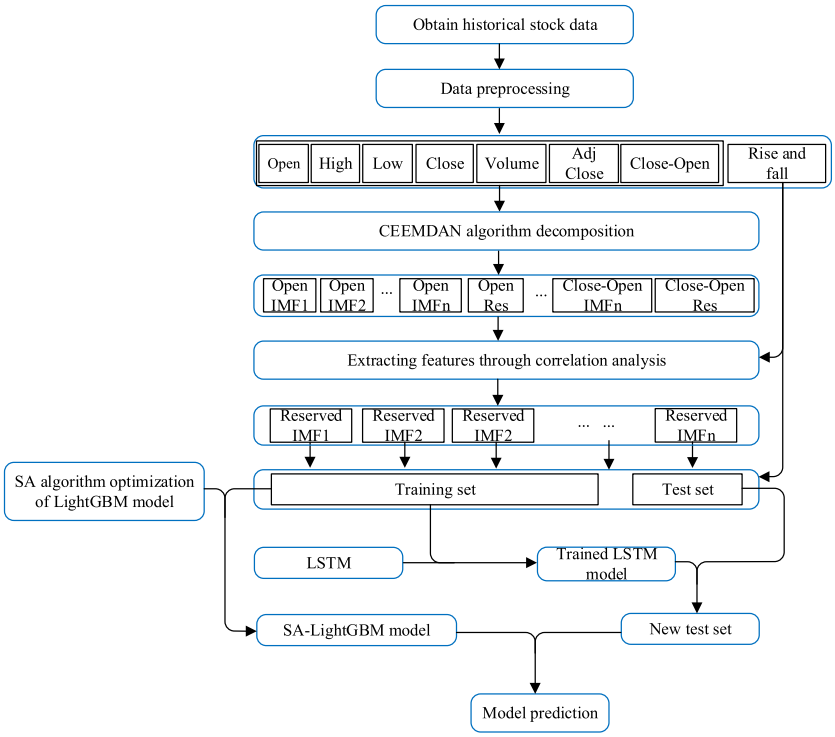

FIGURE 3. Flowchart of the proposed model.

TABLE 2. Descriptive statistical results of stocks.

Stock name	Number of samples	Mean	Min	Max	Std	skewness	kurtosis	J-B value
Salesforce	2518	125.10	36.75	309.96	67.31	0.66	-0.61	222.03
Ali baba	2030	148.96	57.39	317.14	59.89	0.42	-0.72	104.24
Walt Disney	2518	111.49	50.58	201.91	30.58	0.68	0.36	205.85
Electronic Arts	2518	89.991	13.70	148.97	39.36	-0.40	-1.07	188.50
Netflix	2518	239.05	13.14	691.69	174.20	0.58	-0.82	210.20
AMD	2518	33.189	1.62	161.91	38.16	1.18	0.25	591.55

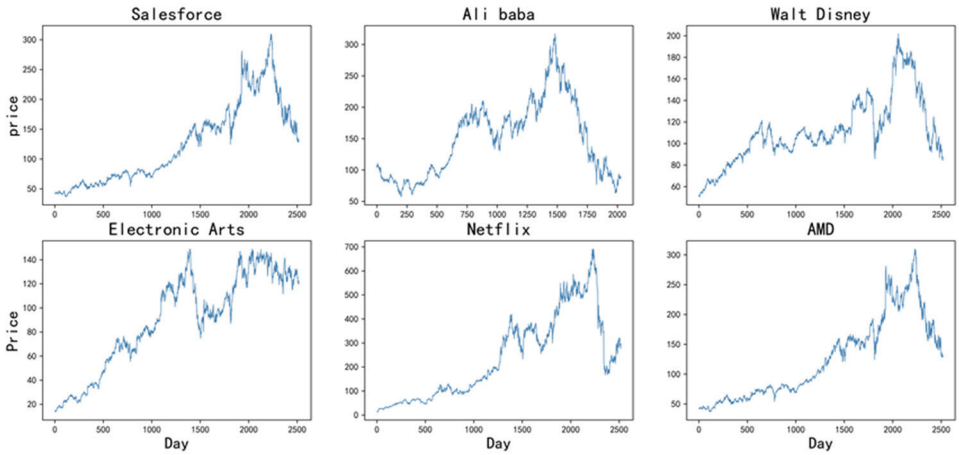

FIGURE 4. The original series of six stock daily closing price.

V. ANALYSIS OF EXPERIMENTAL RESULTS

After the series of modeling work, CEEMDAN-LSTM-SA-LightGBM, a new predictive model has been constructed. Next, this section will validate the forecast performance of the model, including the introduction of evaluation indicators

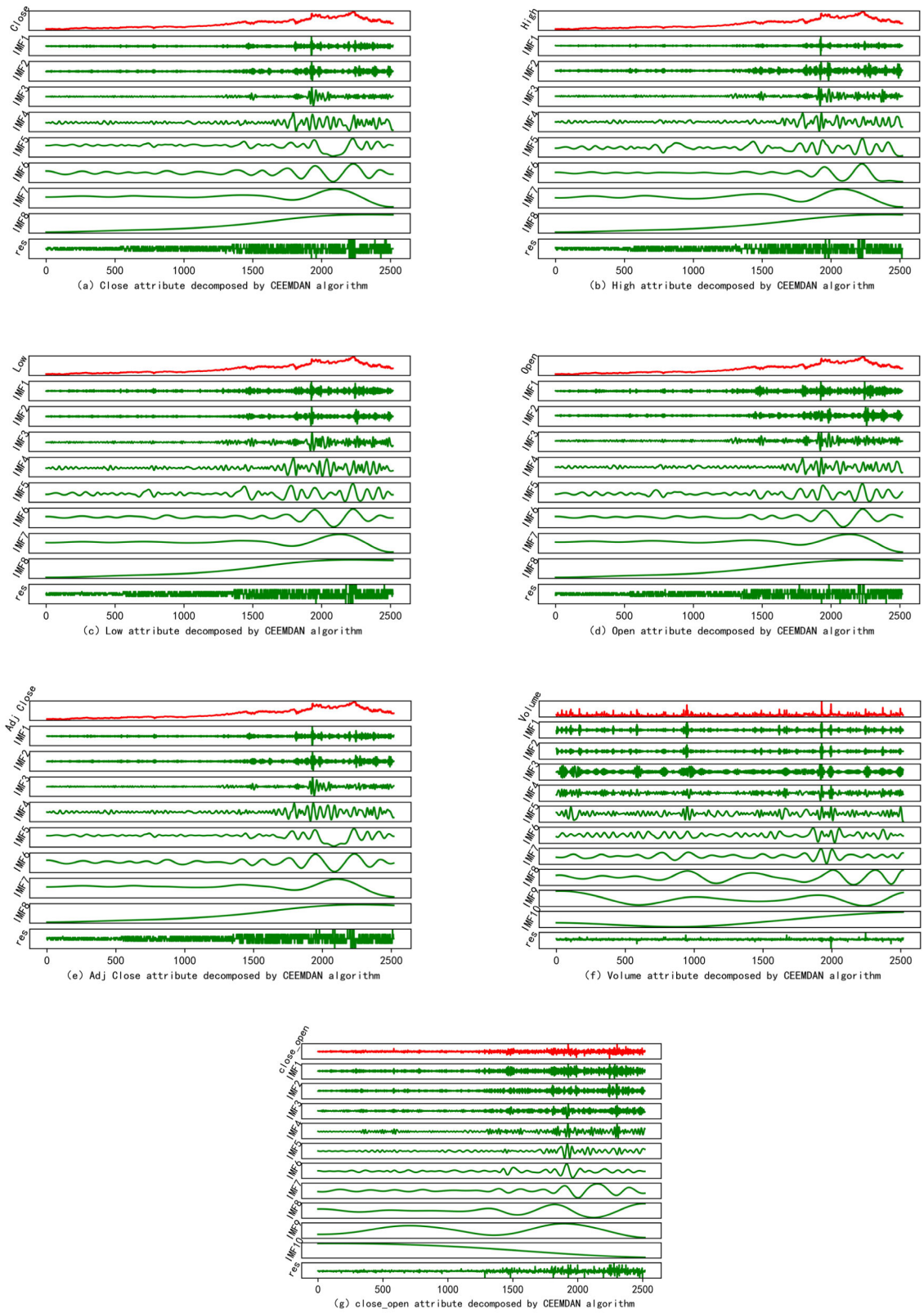

FIGURE 5. Decomposition result of each attribute in Salesforce.

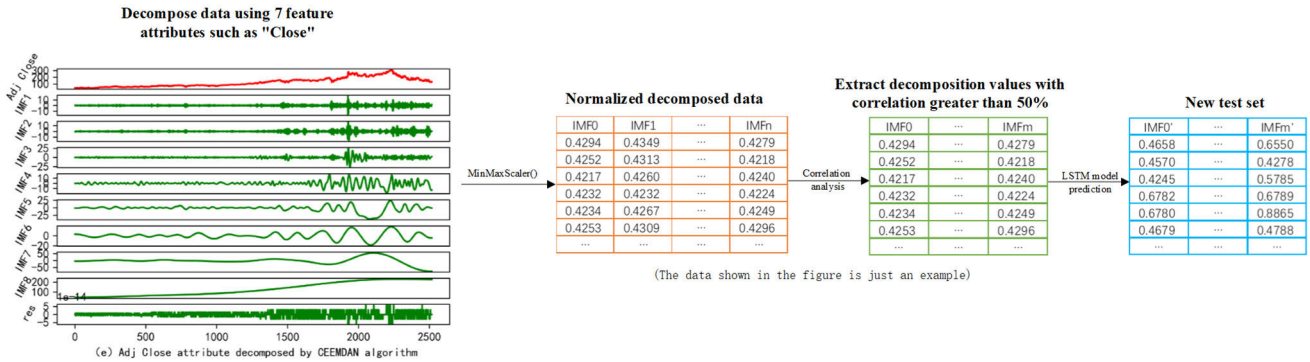

FIGURE 6. Process of establishing new dataset.

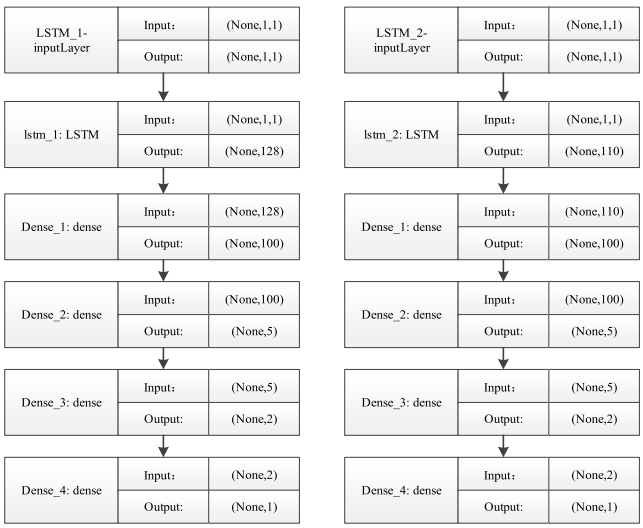

FIGURE 7. The structure of proposed LSTM.

and empirical analysis of the predictive performance with comparative models.

A. EVALUATION INDICATOR

As for model evaluation, this research employs two dimensions of evaluation indicators: fit and accuracy for analysis. The performance evaluation indicators of the stock prediction model are Root Mean Square Error (RMSE) (16), Mean Absolute Error (MAE) (17), and Accuracy (18). The RMSE and MAE indicators are utilized to evaluate the fit of the model.

$$RMSE = \frac{1}{n}\sqrt{\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2} \tag{16}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y^{(i)} - \hat{y}^{(i)}\right| \tag{17}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{18}$$

In the equations above, y and $\hat{y}$ represent the true value and the predicted value respectively. In the accuracy evaluation indicator equation, TP and FP are the correct and incorrect numbers for predicting stock price increases, while TN and FN are the correct and incorrect numbers for predicting stock price decreases.

B. COMPARISON OF EXPERIMENTAL RESULTS

To validate the predictive performance of the hybrid CEEMDAN-LSTM-SA-LightGBM model proposed in this research, a comparative analysis will be conducted with the other 9 popular prediction models. The corresponding names of 10 comparative models are shown in Table 3, and the parameters of the algorithms for comparative analysis are shown in TABLE 4. Fig. 10 shows the prediction results of different models for Salesforce stock.

TABLE 3. The corresponding names of 10 comparative models.

Model number	Model name	Model number	Model name
Model 1	CEEMDAN-LSTM-SA-LightGBM	Model 6	LSTM-LightGBM
Model 2	CEEMDAN-LSTM-LightGBM	Model 7	LSTM-AdaBoost
Model 3	LSTM-SA-LightGBM	Model 8	LSTM-XGBoost
Model 4	LSTM-SA-AdaBoost	Model 9	LSTM
Model 5	LSTM-SA-XGBoost	Model 10	RNN

It is obvious from Fig. 10 that the predicted value of model1 is the closest to the original value, and the prediction effect of the hybrid models are better than the single models. In addition, the hybrid models decomposed by the CEEMDAN algorithm show more excellent prediction effects among all the comparison models. Due to the large number of comparison models, there may be an overlap in the predicted results shown in the figure. Therefore, the next step will be to analyze them through the evaluation indicators.

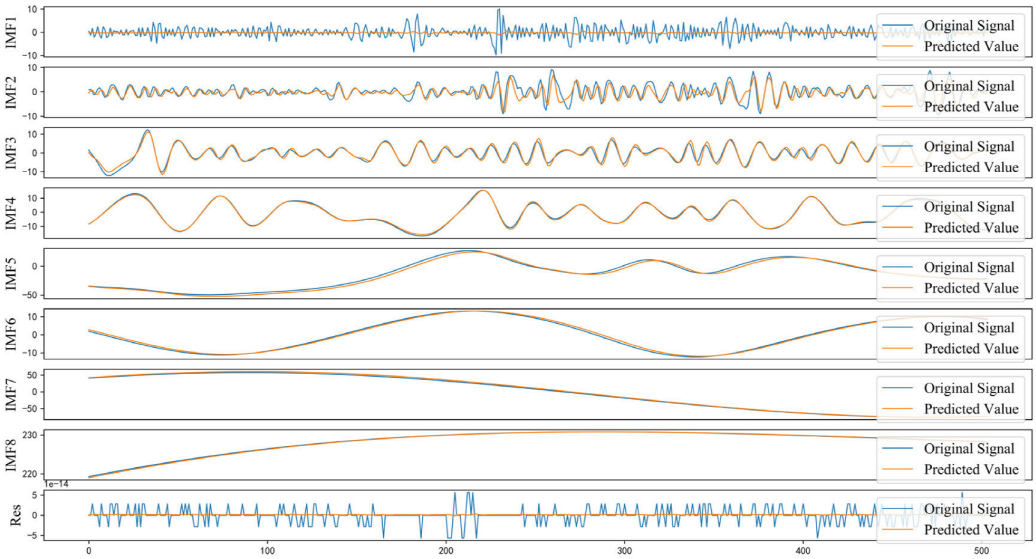

FIGURE 8. Component prediction results after decomposition of the Salesforce's "Close" attribute.

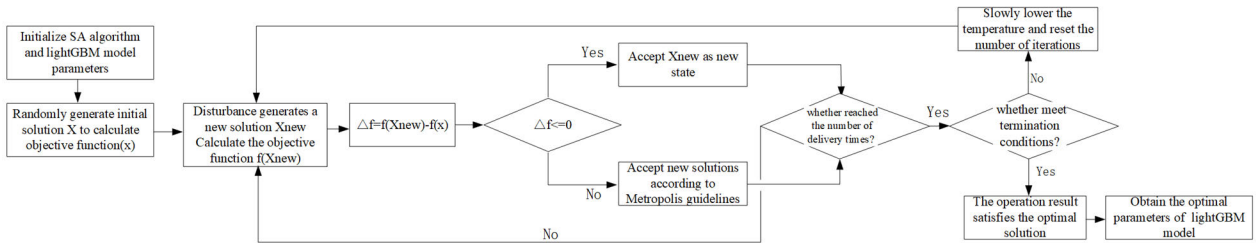

FIGURE 9. Flow chart of parameter optimization.

TABLE 4. Parameters of algorithms used in this work.

Model	Parameter	Value	Model	Parameter	Value
Model 1	n_estimators	quniform(1, 1000)	Model 6、7、8	All	Default
Model 1	max_depth	quniform(1, 30)	Model 9、10	Dropout	0.2
Model 1	learning_rate	loguniform(-5, 1)	Model 9、10	Number of layers	2
Model 1	Validation type	K-fold (cv=5)	Model 9、10	First layer units	32
Model 2	All	Default	Model 9、10	Second layer units	8
Model 3、4、5	n_estimators	quniform(1, 1000)	Model 9、10	Second layer units	8
Model 3、4、5	max_depth	quniform(1, 30)	Model 9、10	batch_size	25
Model 3、4、5	learning_rate	loguniform(-5, 1)	Model 9、10	epochs	10

The bar chart in Fig. 11 shows the average distribution of RMSE, MAE, and accuracy of 10 models after 10 times training sessions. Through comparison, it is very clear to see that the CEEMDAN-LSTM-SA-LightGBM model proposed in this study has the best predictive performance in the three evaluation indicators of RMSE, MAE and Accuracy.

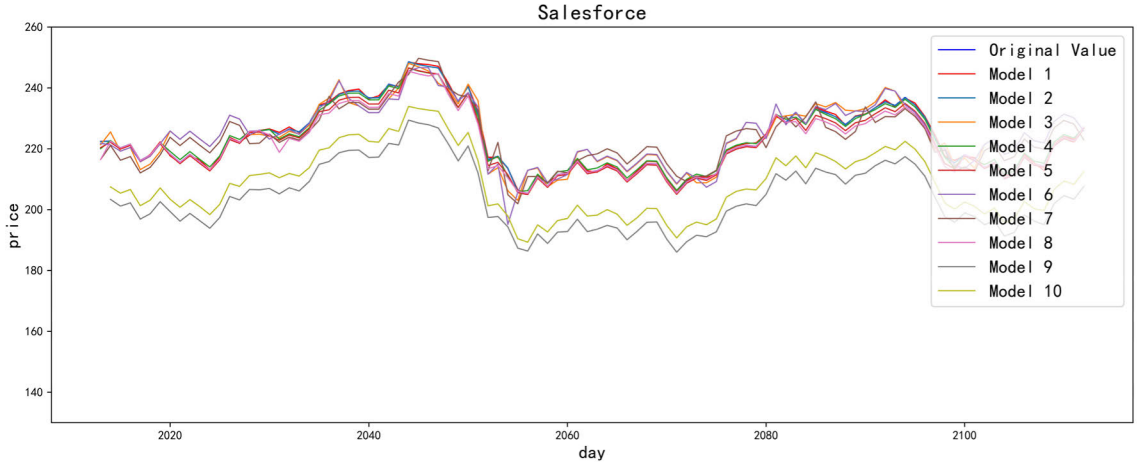

FIGURE 10. Predicted result from the CEEMDAN-LSTM-SA-LightGBM model.

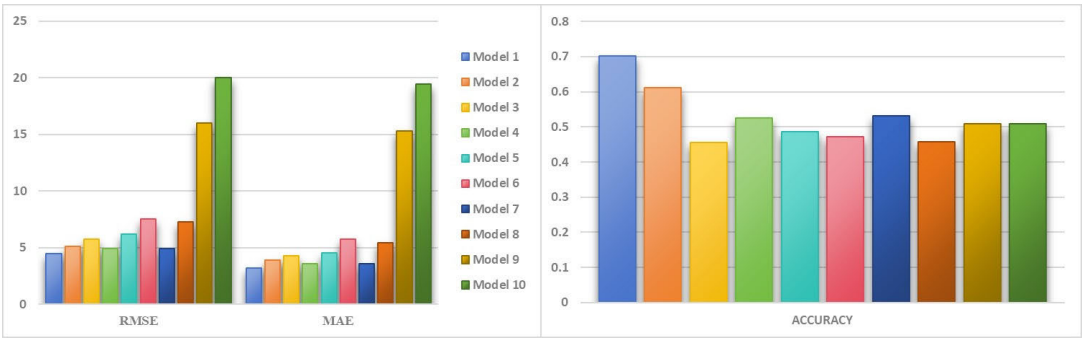

FIGURE 11. Comparison of fit and accuracy of predictions from different models.

TABLE 5. Comparison results of predictions between various models of "Salesforce."

Model number	RMSE	MAE	Accuracy
Model 1	**4.4808**	**3.2172**	**0.7023**
Model 2	5.1207	3.9010	0.6111
Model 3	5.7717	4.3115	0.4563
Model 4	4.9390	3.6178	0.5257
Model 5	6.1690	4.5472	0.4861
Model 6	7.5533	5.7260	0.4722
Model 7	4.9332	3.6079	0.5317
Model 8	7.3103	5.4457	0.4583
Model 9	15.988	15.290	0.5079
Model 10	20.028	19.470	0.5079

The CEEMDAN-LSTM-SA-LightGBM model proposed in this research has the lowest RMSE, MAE, and the highest Accuracy of above 70% in forecasting the rise and fall of "Salesforce" stock.

As seen in TABLE 5, in terms of accuracy evaluation, it is about 9.12% higher than the CEEMDAN-LSTM-LightGBM model and about 20% higher than other models without CEEMDAN decomposition. In terms of fit evaluation, it is about 12.5% and 17.69% lower than the CEEMDAN-LSTM-LightGBM model in the RMSE and MAE indicators respectively, and much lower than other models without CEEMDAN decomposition. This further demonstrates that the CEEMDAN algorithm can effectively eliminate noise, contributing to enhanced prediction accuracy.

For testing and verifying the generalization capability of the model proposed in the research, the CEEMDAN-LSTM-SA-LightGBM model is applied to the prediction of the other five stocks, and compared with the 9 same comparison models. The comparison results are presented in TABLE 6 - TABLE 10.

According to TABLE 6 - TABLE 10, it can be seen that the CEEMDAN-LSTM-SA-LightGBM model consistently shows the lowest RMSE and MAE evaluation indicators in predicting the other five stocks, which indicates that CEEMDAN-LSTM-SA-LightGBM has a higher goodness-of-fit than the other nine models. The model also

TABLE 6. Comparison results of predictions between various models of "Ali baba."

Model number	RMSE	MAE	Accuracy
Model 1	**3.8425**	**2.8162**	**0.7093**
Model 2	5.2504	3.9823	0.6133
Model 3	4.8237	3.5385	0.4532
Model 4	5.2208	3.2865	0.4753
Model 5	6.3927	4.3911	0.3891
Model 6	6.1965	4.7582	0.4187
Model 7	5.2371	3.3054	0.4384
Model 8	6.5802	4.8739	0.4014
Model 9	7.1225	6.1212	0.4334
Model 10	11.325	10.561	0.4334

TABLE 7. Comparison results of predictions between various models of "Walt Disney."

Model number	RMSE	MAE	Accuracy
Model 1	**2.3366**	**1.7234**	**0.738**
Model 2	3.0982	2.2726	0.6567
Model 3	3.2203	2.3364	0.4543
Model 4	2.8776	2.0481	0.4722
Model 5	3.4936	2.5961	0.4464
Model 6	4.5702	3.0841	0.4126
Model 7	3.0553	2.1278	0.4781
Model 8	4.0317	2.9863	0.4246
Model 9	3.6285	2.9077	0.5357
Model 10	4.2227	3.5345	0.5357

TABLE 8. Comparison results of predictions between various models of "Electronic Arts."

Model number	RMSE	MAE	Accuracy
Model 1	**1.86**	**1.4122**	**0.6706**
Model 2	2.3473	1.7768	0.5753
Model 3	2.7428	2.1604	0.4583
Model 4	2.1349	1.6275	0.4702
Model 5	2.9331	2.2803	0.4365
Model 6	3.529	2.7964	0.4384
Model 7	2.1399	1.6516	0.4841
Model 8	2.9762	2.3098	0.4146
Model 9	2.3082	1.8036	0.49
Model 10	2.3168	1.8256	0.49

TABLE 9. Comparison results of predictions between various models of "Netflix."

Model number	RMSE	MAE	Accuracy
Model 1	**11.24**	**6.6153**	**0.75992**
Model 2	13	8.6454	0.6706
Model 3	14.89	9.4475	0.4603
Model 4	13.089	8.1252	0.4781
Model 5	15.168	9.503	0.494
Model 6	17.652	11.347	0.4702
Model 7	13.31	8.2818	0.4821
Model 8	16.676	10.714	0.4742
Model 9	28.143	25.896	0.5079
Model 10	35.317	33.585	0.5079

demonstrates excellent performance in the evaluation indicator of Accuracy. It has a prediction accuracy of over 67% for all five stocks. Compared to other models, especially the single LSTM and RNN prediction models, the prediction accuracy of the hybrid model proposed in this study has been greatly improved. Overall, the performance of the CEEMDAN-LSTM-SA-LightGBM model in predicting stock price is relatively stable.

To sum up, in the experimental validation of six stocks, including "Salesforce", "Ali Baba", "Walt Disney", "Electronic Arts", "Netflix", and "AMD", it is evident that the CEEMDAN-LSTM-SA-LightGBM model outperforms the CEEMDAN-LSTM-LightGBM model, LSTM-SA-LightGBM model, and other comparative models in predicting stock price and fluctuations. The prediction validation involving six stocks demonstrates that the

TABLE 10. Comparison results of predictions between various models of "AMD."

Model number	RMSE	MAE	Accuracy
Model 1	**3.0851**	**2.2318**	**0.6845**
Model 2	3.1401	2.4676	0.625
Model 3	3.5879	2.7464	0.4583
Model 4	3.4599	2.5256	0.4801
Model 5	4.0828	3.1119	0.4861
Model 6	4.1306	3.0444	0.494
Model 7	3.5039	2.5554	0.5178
Model 8	4.2167	3.1504	0.4742
Model 9	3.5833	2.7182	0.5178
Model 10	3.4782	2.5958	0.5158

CEEMDAN-LSTM-SA-LightGBM model proposed in the study exhibits advantages in both fitting and accuracy.

VI. CONCLUSION

This study proposes a novel CEEMDAN-LSTM-SA-LightGBM hybrid model based on the CEEMDAN algorithm, double-layer LSTM, and simulated annealing optimized LightGBM model to improve the fitting ability and accuracy of stock price prediction. The effectiveness and stability of the model have been proven through multiple verifications and comparisons. The main conclusions are as follows:

(1) This framework is based on the CEEMDAN algorithm which can reduce the noise in the original signals to significantly improve the predictive performance, but requires a relatively long time in the decomposition process.

(2) The method of feature selection can effectively save computational costs.

(3) LSTM is a mature and excellent prediction model, it has shown stable and reliable performance in stock price prediction.

(4) The ensemble learning algorithm has good stability, and the prediction performance of the method combining the ensemble learning algorithm with the neural network is better than that of the other comparative hybrid models and single models.

The approach proposed in this study can simultaneously enhance both the fit and accuracy of stock price fluctuations prediction, but there are still challenges and limitations of this research, mainly as follows:

- The selection of parameters such as the number of sliding window sizes in neural networks is still somewhat subjective;
- The double-layer LSTM elements may not be able to fully explore the deep variation pattern information contained in nonlinear complex exponential sequences, so further research is needed to optimize model parameters;
- In feature extraction, further comparison and analysis can be conducted through different proportions of correlation.
- Lack of comprehensive stock information.

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